NEHA SUNIL

nsunil@mit.edu
Portfolio: nehasunil.com
Cambridge, MA
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EDUCATION

MIT
PhD in Mechanical Engineering
Minor: Machine Learning
2021 - 2025 | Cambridge, MA
GPA: 4.9/5.0

MIT
MS in Mechanical Engineering
2019 - 2021 | Cambridge, MA
GPA: 5.0/5.0

CALTECH
BS in Mechanical Engineering
2015 - 2019 | Pasadena, CA
GPA: 3.9/4.0

COURSEWORK

GRADUATE

Deep Learning
Bayesian Inference and Modeling
Intelligent Robotic Manipulation
Underactuated Robotics
Visual Navigation
Computer Vision
Analysis/Design of Feedback Control
Precision Product Design
Dynamics

UNDERGRADUATE

Capstone Design Contest
Robotics & Autonomy
Experimental Robotics
Multidisciplinary Systems Eng.
Mechatronics
Biotechnology Lab
Microfabrication Lab
Information and Logic
Experiments & Modeling in MechE
(Teaching Asst 2x)

SKILLS

PROGRAMMING

Python • PyTorch • Java • C • C++
Matlab • R • Mathematica
HTML • VB • LaTeX

ROBOTICS

ROS • Drake • Simulink
OpenCV • Blender
Raspberry Pi • Arduino
UR5 • ABB YuMi • GelSight

ENGINEERING

CAD: Solidworks • Onshape
CFD/FEA: Ansys • COMSOL

RESEARCH

MIT MCube Lab, PhD Research Assistant | 2019 - 2025
Advisor: Alberto Rodriguez
Built reactive systems for deformable object manipulation (e.g., cables and cloth) using visual and tactile sensing. Developed generalizable control strategies driven by local feedback

CALTECH Undergraduate Researcher | 2018 - 2019
Advisors: Chiara Daraio and Aaron Ames
Designed a flexible and inexpensive pressure sensor to determine the real-time center of pressure for walking robots and prostheses

STANFORD Student Researcher | 2014
Advisor: Anshul Kundaje
Developed a library in R to choose DNA regions for CRISPR-Cas9 gene editing technology

INDUSTRY

VERB SURGICAL Mechanical Engineering Intern | Summer 2018
Google and J&J robotic surgery partnership | Mountain View, CA
Worked with team interfacing between arm and surgical tool
Experience in design, sensors, and controls

NIMA LABS Mechanical Engineering Intern | Summers 2016, 2017
Portable food allergen sensor startup | San Francisco, CA

- 2016: Tested multi-channel version of consumer device and isolated key variables affecting chemistry development and camera readings
- 2017: Redesigned multi-channel device from scratch. Created manufacturing and assembly drawings and worked with vendors

SELECTED PUBLICATIONS

[1] Y. She*, S. Wang*, S. Dong*, N. Sunil, A. Rodriguez, and E. Adelson. "Cable Manipulation with a Tactile Reactive Gripper". In: *The International Journal of Robotics Research (IJRR)* (2021). **RSS'20 Best Paper Award Finalist**.

[2] N. Sunil*, S. Wang*, Y. She, E. Adelson, and A. Rodriguez. "Visuotactile Affordances for Cloth Manipulation with Local Control". In: *CoRL* (2022).

[3] N. Sunil*, M. Tippur*, A. Saumell, E. Adelson, and A. Rodriguez. "Reactive In-Air Clothing Manipulation with Confidence-Aware Dense Correspondence and Visuotactile Affordance". In: *CoRL* (2025). **Featured Oral Presentation**.

More projects and details available at **nehasunil.com**

ACHIEVEMENTS & SERVICE

2022	Mentored Visiting Student Arnau Saumell
2020	RSS Best Paper Award Finalist
2019	Co-founded Graduate Women in Robotics Community (GWiRC) at MIT
2019	NSF Graduate Research Fellowship
2019	MIT Linden/Wong Mechanical Engineering Fellowship
2019	Paul & Daisy Soros Fellowship Finalist
2019	Caltech Mechanical Engineering Award
2018	Tau Beta Pi Engineering Honor Society
2011	Certified Yoga Instructor